Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/25/2007
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Increasing the Efficiency of Exercise Countermeasures for Bone Loss		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sbloom@tamu.edu	**Fax:**	FY 979-862-1692
PI Organization Type:	UNIVERSITY	**Phone:**	979-845-2871
Organization Name:	Texas A&M University		
PI Address 1:	Department of Health & Kinesiology		
PI Address 2:	400 Harvey Mitchell Pkwy, Suite 300		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Hogan, Harry (Texas A&M University)		
Grant/Contract No.:	NCC 9-58-BL00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: The over-arching goal of the proposed research is the test the hypothesis that astronaut exercise time devoted to mitigation of bone loss can be reduced if combined with an anti-resorptive pharmacological agent. This combined strategy is likely to be more effective than exercise or alendronate alone at achieving reversal of the frequently observed suppressed bone formation and of the accelerated resorption observed with simulated microgravity. Importantly, skeletal muscle mass, function, and anabolic markers will also be assessed to assure that any reductions in exercise time do not sacrifice benefits to muscle. The original aims of the project were to 1) test two different modes of resistance training (electrically stimulated muscle contractions versus voluntary resistance exercise) for their effects on mitigating bone and muscle decrements during 28 days of hindlimb unloading (HU) in adult rats; 2) to test different doses of alendronate during 28 days HU for maximal efficacy in suppressing bone resorption, and then to combine alendronate dosing with that training paradigm judged most successful to test for additive or synergistic effects; and 3) to determine whether lower “doses” of exercise, when combined with this anti-resorptive agent, can provide an acceptable level of prevention against unloading-induced decrements in bone and muscle parameters. These integrated experiments will provide fundamental and unique information about the biological and mechanical interactions of these two integrated tissues in response to the anabolic effects of exercise and the anti-catabolic effects of alendronate. The outcomes of this project are easily translatable to human studies using appropriate ground- and flight-based test beds. Prior Progress: The first year of this project was largely devoted to the development of our flywheel device for voluntary resistance training in HU rats and to modifications of and preliminary studies with our muscle stimulation system, which allows for precisely controlled muscle contraction regimens of the rat’s lower leg and also measurement of muscle strength by measuring torque produced at the ankle. We became expert in the operant conditioning required for the voluntary resistance training, and attempted several pilot studies. Experiments with our first protocol of simulated resistance training (with muscle stimulation) were completed in Year 2, were remarkably successful in preventing the unloading-induced bone loss at the proximal tibia usually observed after HU. Bone strain data suggested that bone strain responses to imposed muscle contraction changed with duration of disuse (HU). Markers of muscle function, surprisingly, were not improved; assays for muscle anabolic markers yielded equivocal results. Progress in Year 3: A revision of the simulated resistance training protocol was completed in hopes of maintaining the positive impact on bone mass and strength and providing some mitigation of loss of skeletal muscle function and mass. The primary changes in the training paradigm were 1) decreasing the intensity of muscle contractions and 2) changing the range of motion allowed at the ankle during lower leg contractions, to achieve a greater stretch on the posterior crural muscle group. This training regimen proved even more effective for mitigating negative bone changes, now impacting positively on bone geometry and mineral content at the mid-shaft tibia as well as at the proximal site; however, muscle function and mass was minimally affected. To our knowledge, this is the first countermeasure regimen to demonstrate an absolute gain in bone mass during simulated microgravity. Parallel strain gage studies during simulated resistance training are providing unique information about the nature of the strain stimulus experienced by bone during isometric and eccentric muscle contractions. Achilles tendon and knee ligaments from these animals have been collected in hopes of completing collaborative studies to define the efficacy of this training for those soft connective tissues. Plans for Year 4: One final modification of our muscle stimulation protocol (for simulated resistance training during HU) is currently being tested, in hopes of demonstrating a positive impact on muscle outcomes. Tissues collected over the previous 6 months will be shipped to co-investigator KM Baldwin for detailed analyses of muscle anabolic markers, which may be more sensitive indicators than our muscle function testing. Hence, Specific Aim 1 animal experiments are virtually complete and several manuscripts are now in preparation detailing those results. Preliminary studies on optimizing the dose of alendronate for studies will be completed by early July 2007, allowing us to perform the key experiment for Specific Aim 2: testing for additive and synergistic effects of alendronate (anti-catabolic) and simulated resistance training (anabolic) in mitigating bone loss with unloading. By fall of 2007 we will test the key hypotheses behind Specific Aim 3: can we reduce the volume of training required to produce mitigation of bone loss when combined with alendronate? More frequent shipments of muscle samples to Co-I Baldwin will allow muscle anabolic marker assay results to keep pace with the PI lab’s results. Specific Aim 3 experiments should be completed by early 2008, simultaneous with data analysis and manuscript preparation of preceding work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our detailed dose-response studies of the bone response to resistance training applied as a countermeasure during a period of simulated microgravity will help define the training stimulus required for maintaining bone health during prolonged periods of bed rest in clinical patients or others restricted to bed rest. Subsequent trials will test the efficacy of alendronate, an FDA-approved anti-resorptive agent commmonly prescribed for osteoporotic patients, versus resistance training versus combined alendronate and exercise in maintaining bone health during a period of unloading. These studies will answer whether there is an additive or synergistic effect of these pharmacological and physiological approaches. Given the numbers of Americans currently taking alendronate and other closely related bisphosphonate agents who also exercise, this could prove to be very useful data that could be used effectively to predict the human response to the combined effects of exercise and bisphosphonate treatment. Finally, our experiments testing various durations of exercise in combination with alendronate will quantify the minimum volume of exercise required to achieve the desired outcome of maintaining bone mass during prolonged periods of non-weightbearing when combined with this anti-resorptive treatment. This has important implications for rehabilitation specialists, physical therapists, and any clinicians treating bedridden patients.
	Progress continues on identifying an optimal resistance training paradigm for mitigating bone and muscle loss in a rodent hindlimb unloading model. 1. Completion by K. Baldwin's lab (Co-I) of gene expression assays of gastrocnemius muscle samples from our pilot study (in Year 2) to test the effectiveness of our muscle stimulation protocol in normally weightbearing rats [adult male rats were exposed to six training sessions using fine percutaneous needles over 2 weeks (4 sets of 10 contractions at 120% peak isometric torque, 3 days/week)]. 2. Completion of second full set of hindlimb unloading (HU) + stimulated eccentric muscle contraction (STIM) experiments mimicking resistance training, using a modified training protocol vs. previous experiment (Year 2). The

Task Progress:	goal was to achieve same prevention of bone loss with HU as observed in previous trials, but achieve some mitigation of muscle atrophy and loss of muscle function, which was not accomplished in the first study. Outcomes: in vivo muscle functional properties, pQCT (pre-post) measures of vBMD and cross-sectional geometry on proximal and mid-shaft tibia, RPC testing on proximal tibia for mechanical integrity, muscle weights. Frozen medial gastrocnemius samples were sent to K. Baldwin's laboratory for gene expression assays. Bone samples are stored for histomorphometric studies of bone formation rates and cancellous bone microarchitecture. 3. Follow-up studies on characterization of bone strain during stimulated muscle contractions: a companion study with the HU + STIM experiments described above to characterize bone strains incurred during these stimulated muscle contractions. Last year we collected data on 3 rats on Days 0 , 7 and 21 d of HU; this time around, using the revised training protocol, we allocated 7 rats each to testing on Days 0 and 21, in order to achieve maximal statistical power. 4. Development of our other training paradigm (voluntary resistance exercise on a flywheel device): continuing difficulties with optimizing the rat-machine interface forced a decision to divert resources back to the simulated resistance training using stimulated muscle contractions. In addition, this latter paradigm allows us to more precisely titrate the exercise stimulus when combined with alendronate treatments, which is important to the final set of experiments planned in Year 4. 5. Investigator meetings with K.M. Baldwin were held at the American College of Sports Medicine meeting in Denver, May 2006 and at the NASA/NSBRI Retreat in February, 2007; frequent discussions about data and experimental design improvements occur via e-mail communications. Co-Investigator Harry Hogan and PI Bloomfield hold joint lab meetings throughout the year, maximizing communication flow and collaboration on these studies.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Nilsson M, Swift JM, Stallone JL, Baek K, Bloomfield SA. "Reduced mechanical loading does not diminish the anabolic response of bone to intermittent pth." 36th Sun Valley Skeletal Biology Workshop, Sun Valley, ID, July 30 - August 2, 2006. J Musculoskel Neuron Interact. 2006 Oct-Dec;6(4):397. , Oct-2006
Abstracts for Journals and Proceedings	Swift JM, von Walden F, Elman T, Fluckey JD, Bloomfield SA. "Assessing the efficacy of multiple dosing regimens of a beta-agonist agent for bone and muscle loss during 28-day hindlimb unloading." 36th Sun Valley Skeletal Biology Workshop, Sun Valley, ID, July 30 - August 2, 2006. J Musculoskel Neuron Interact. 2006 Oct-Dec;6(4):399. , Oct-2006
Abstracts for Journals and Proceedings	Alcorn JD, Vyvial BA, Bloomfield SA, Hogan HA. "Exercise improves the strength of bone in the proximal tibia of hindlimb unloaded mature rats." 5th World Congress of Biomechanics, Munich, Germany, July 29-August 4, 2006. J Biomech. 2006;39(1 Suppl):S13. , Jul-2006
Abstracts for Journals and Proceedings	Hogan HA, Alcorn JD, Vyvial BA, Bloomfield SA. "Simulated resistive exercise benefits bone in hindlimb unloaded rats." 28th Annual Meeting, American Society of Bone and Mineral Research, Philadelphia, PA, September 15-19, 2006. J Bone Miner Res. 2006;21(1 Suppl):S151. , Sep-2006
Articles in Peer-reviewed Journals	Allen MR, Hogan HA, Bloomfield SA. "Differential bone and muscle recovery following hindlimb unloading in skeletally mature male rats." J Musculoskelet Neuronal Interact. 2006 Jul-Sep;6(3):217-25. PMID: 17142941 , Jul-2006
Articles in Peer-reviewed Journals	Bloomfield SA. "Does altered blood flow to bone in microgravity impact on mechanotransduction?" J Musculoskelet Neuronal Interact. 2006 Oct-Dec;6(4):324-6. PMID: 17185808 , Oct-2006
Articles in Peer-reviewed Journals	Baek K, Barlow AA, Allen MR, Bloomfield SA. "Food restriction and simulated microgravity: effects on bone and serum leptin." J Nutrition. Submitted, 2007. , May-2007
Awards	Baldwin KM. "Honor Award, Environmental & Exercise Physiology Section, American Physiological Society, April 2006." Apr-2006
Awards	Hogan HA. "Claude L. Wilson Award, American Society of Mechanical Engineering South Texas Section, April 2006." Apr-2006
Dissertations and Theses	Alcorn JD. "Osteogenic effect of electric muscle stimulation as a countermeasure during hindlimb unloading." Master's thesis, Dept. of Mechanical Engineering, Texas A&M University, May 2006. , May-2006
Dissertations and Theses	Vyvial BA. "Characterizing strain in the rat proximal tibia during in vivo electrical muscle stimulation. " Master's thesis, Dept of Mechanical Engineering, Texas A&M University, May 2006. , May-2006